

Elements of Position

II. Without an Angle

Two moves, the pair they force, and two mixes that must not be crushed

Justin Erholtz

12 September 2026

The operators do not contain the joint.

A note. Part I named the pair on the integer and the lock at $n = 1$. This part walks. Identities of the pair are citations of Part I. A gap, a word, a mean taken over one run, are reported from a named table. The recorded run is `kernels/walk/runs/2026-09-24_h0.002_n200000/`: $h = 0.002$, 200,000 steps, $H_0 = 1571$, 127 hits. That summary is byte-identical to the 19 September record. A different kick writes a new table. It does not revise an identity.

A note on names. Shear T_h , bind N , step, hit, heading u_m . Outer point $P_m = mu_m$. Inverse point $Q_m = u_m/m$. Half-turn mix — $h = \tan(\pi/n)$. Vertex mix — $h = \tan(2\pi/n)$. These two mixes are not one sentence.

Contents

1	The operators	1
2	The pair is already proved	2
3	Four points	2
4	The count pays with two integers	2
5	Two mixes	3
6	What stands	3

The operators

Start on the diameter: $(1, 0)$.

Definition 1 (Shear). $T_h(x, y) = (x - hy, y + hx)$ for $h > 0$.

Definition 2 (Bind). $N(x, y) = (x, y)/\sqrt{x^2 + y^2}$.

Definition 3 (Step). $\text{step} = N \circ T_h$.

No angle is an argument of either move. Composed, one step is a rotation by $\arctan h$. That identity is classical (Volder, 1959). It is never used inside the walk. It is used only to name a comparison after a table exists.

Definition 4 (Hit). A hit is a step at which the heading's second coordinate changes sign.

On the recorded run the first hit is step $H_0 = 1571$, heading $(-0.99999992, -0.00040316)$. The same first crossing, at the same h , is what the continuous-clock reading counts as its half-turn H . Two detectors, one event.

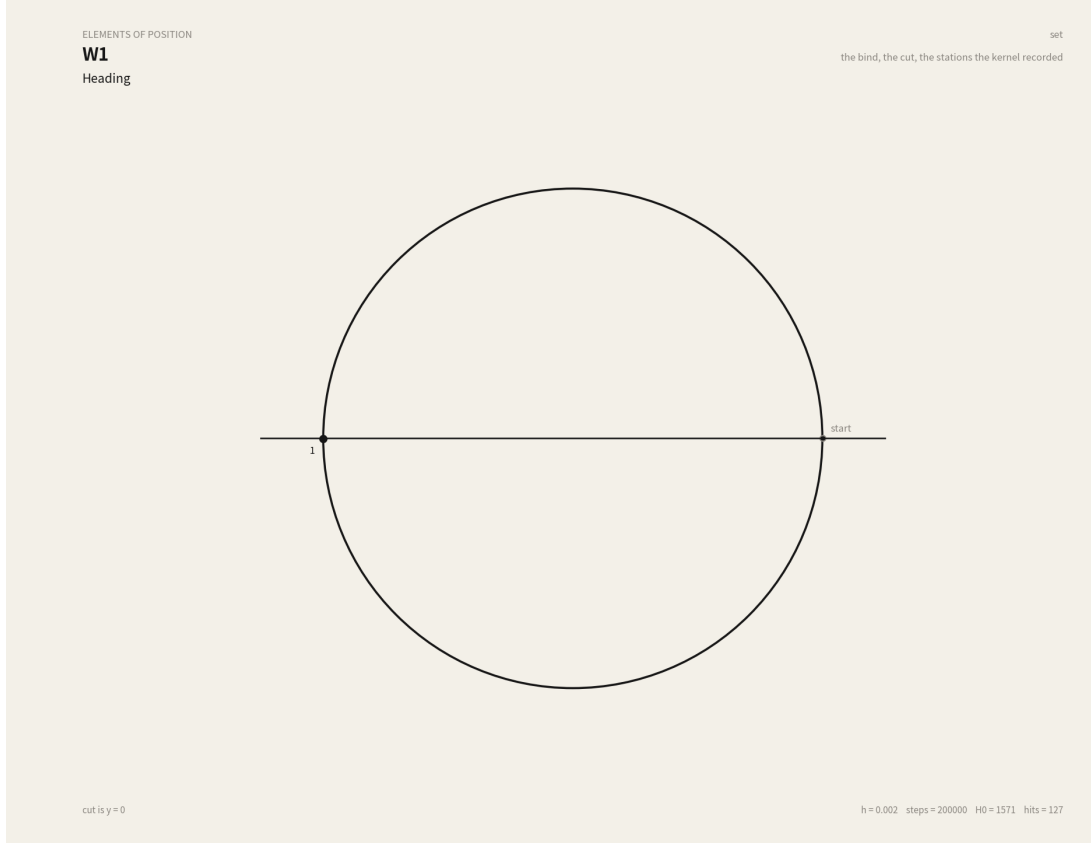


Figure 1: Sheet W1. The bind, the cut, the stations the kernel recorded.

The pair is already proved

Once the hit index m is named, Part I applies with $n = m$.

$$r \cdot \frac{1}{r} = 1, \quad \text{GM} = 1, \quad \tau(m) = \frac{(m-1)^2}{2m}.$$

$P_m = mu_m$ and $Q_m = u_m/m$ are that pair read as points on the recorded ray. Q_m is the inversion of P_m through the bind: $|P_m|^2 = m^2$, so $P_m/|P_m|^2 = Q_m$.

Four points

Proposition 1. *For every $m > 1$, $(1/m, m; 1, -1) = -1$.*

The four points are the pair and the two marks the diameter already fixed. This is the classical involution $z \mapsto 1/z$ on the line.

The circle on diameter OP_m meets the unit circle at points whose foot on the ray of u_m is exactly Q_m . No idealization. Checked on the recorded headings to machine precision.

The count pays with two integers

After the first hit the recorded run writes only two gap values: 1570 and 1571. On 127 hits, 1571 occurs 102 times and 1570 occurs 25 times. Mean gap times kick: 3.14160630.

A named comparison after the table exists:

$$\lambda := \frac{\pi}{\arctan h} = 1570.798421, \quad \{\lambda\} = 0.798421.$$

The walk cannot land on one length, so it writes both neighbors. This is the discrete case of the

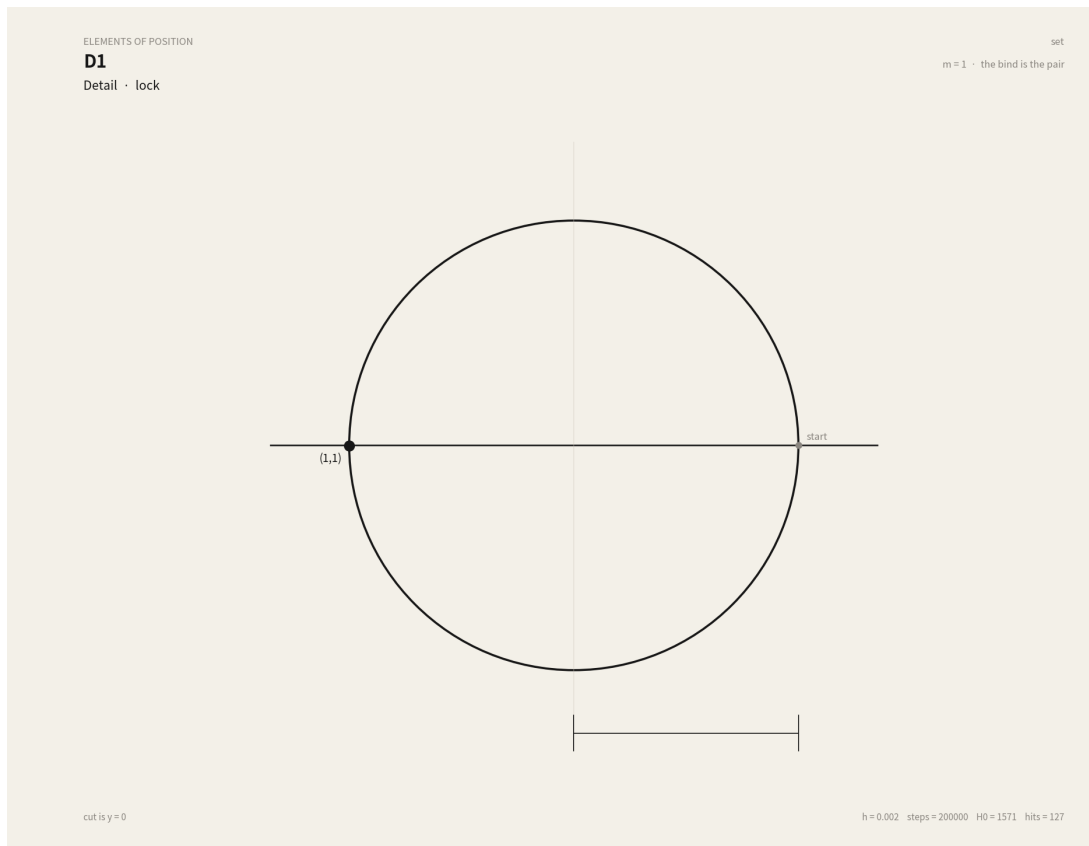


Figure 2: Sheet D1. The lock. Pair (1,1).

three-distance theorem.

Two mixes

Observation 1 (Half-turn mix). Choose $h = \tan(\pi/n)$ for an integer $n \geq 3$. Then $\lambda = n$ exactly, and every gap after the first hit equals n . Checked here: $n = 3, 4, 5, 6, 8$, one letter after the first hit. At $n = 4$, $h = 1$, the square lands.

Observation 2 (Vertex mix). Choose $h = \tan(2\pi/n)$. One elementary step is the central angle of the regular n -gon. For $n \geq 5$ that step is legal. For $n = 4$ the tangent is a pole. For $n = 3$ the tangent of $2\pi/3$ is negative; it is not this series' positive kick.

A hit is a half-turn. A vertex step is $2\pi/n$. Steps per hit are therefore about $n/2$, not n . At the pentagon vertex mix the recorded gaps are $\{2, 3\}$, not $\{5\}$. The vertices are still the pentagon: five steps from $(1, 0)$ close with error 3.4×10^{-17} , sides equal to $2 \sin(\pi/5)$ with error 0. Marriage of the two kernels (Part IV) uses the vertex mix. Exact closure of the gap word uses the half-turn mix. They must not share a sentence that says “at $n = 5$ the walk lands.”

What stands

Forced, once the operators are granted: the step with no angle in either move; a hit as a sign change; the pair and the leftover, by Part I; P_m and Q_m , and Q as inversion of P ; the circle on OP with foot Q , at every residual.

Reported from the named table: two gap integers; mean gap times kick next to π ; $H_0 = 1571$.

Not granted: an angle as an argument of the step; a landing on the cut; the two mixes crushed into one kick; a spiral treated as a different object from the hit plan.

Where is 1? It is still the lock. The operators met themselves. They did not land.

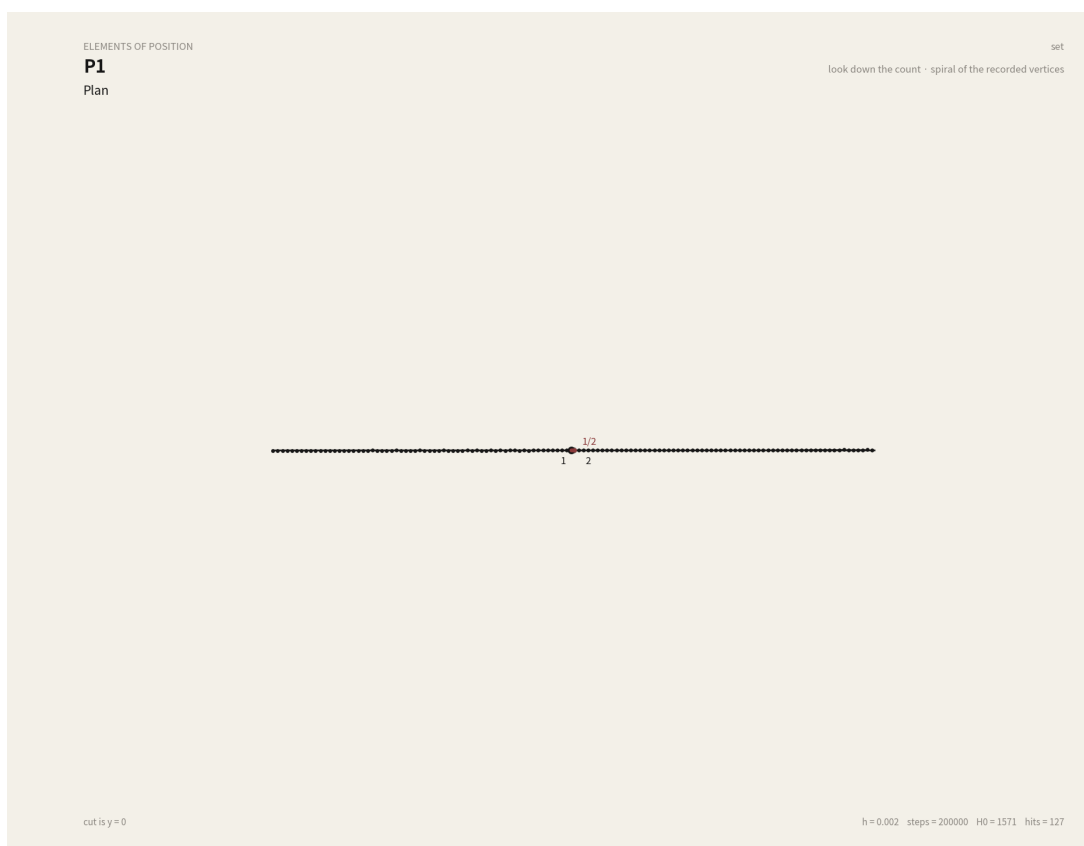


Figure 3: Sheet P1. Plan of the recorded hits. Outer locus in ink; inverse in the accent. Hits sit on the cut, so the plan of P_m is nearly a line. A spiral appears only when every step is drawn with radius growing as $k/2$. That is a later sampling of the same walk, not a second object.

Figure 4: Sheet D6. Circle on OP meeting the bind. The foot is Q .



Figure 5: Sheet SCH1. Two gap integers on the recorded run.